

A Continuous-Time ECM–OCV Model for Smartphone Battery Time-to-Empty Prediction with Constant-Power Load and Debounced Cutoff

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Abstract—Predicting smartphone battery Time-to-Empty (TTE) under realistic workloads is challenging because power demand, electrochemical state, temperature, and aging all interact. We present a continuous-time framework that couples a second-order Thevenin equivalent-circuit model (ECM) and an Open-Circuit Voltage to State-of-Charge (OCV–SOC) lookup with a system-level load model aggregating screen, CPU, cellular, WiFi, GPS and background contributions. A constant-power load constraint is enforced through an implicit per-step current solve, and a debounced cutoff stabilises the shutdown decision. Capacity and resistance temperature dependencies are calibrated from public battery datasets, and component power coefficients are estimated from Android Debug Bridge (ADB) traces of a 4724 mAh device. Across four representative usage profiles at 25 °C the predicted TTE spans 39.45 h (idle), 9.06 h (video), 5.34 h (gaming) and 4.08 h (mixed). Sensitivity analysis identifies screen brightness, network activity, CPU load and baseline draw as dominant TTE drivers, while low-power peripherals contribute marginally. The framework supports user- and OS-level battery-life recommendations and is portable to other consumer devices.

Index Terms—Lithium-ion battery, smartphone, time-to-empty prediction, equivalent circuit model, OCV–SOC, sensitivity analysis, consumer electronics.

I. INTRODUCTION

Smartphones are dominant consumer electronics platforms, and battery life remains one of the most user-visible quality attributes. Yet under nominally similar usage the runtime often varies by hours, because the battery is simultaneously a load-driven electrical system, a thermally coupled cell, and a slowly aging electrochemical device. Prior work has addressed each piece in isolation: equivalent circuit models (ECMs) capture short-term electrical dynamics [1], OCV–SOC characterisation provides voltage–state mappings [2], temperature dependencies follow Arrhenius-type laws [3], and aging models capture cycle-induced fade [4]. Component-level power profiling on Android has been studied since PowerTutor [6].

What is still missing for a consumer-electronics design context is a single, reproducible continuous-time pipeline that (i) takes a workload trace as input, (ii) solves the implicit constant-power coupling between terminal voltage and

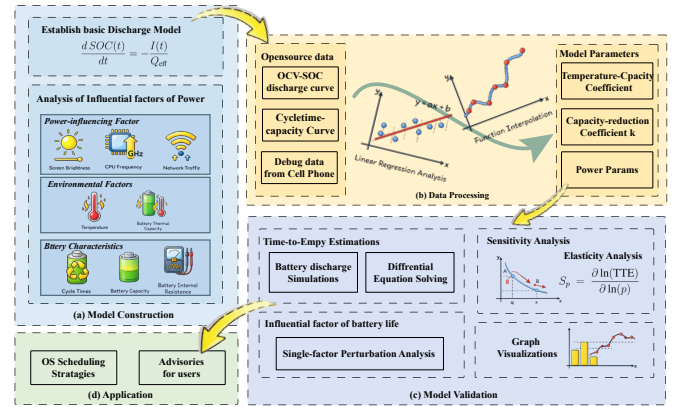


Fig. 1: Modeling and calibration workflow: data acquisition, parameter fitting, scenario simulation, and sensitivity analysis.

battery current, and (iii) returns Time-to-Empty (TTE) and component-resolved drain attributions consistent with both bench data and on-device measurements. This paper contributes such a pipeline:

- A coupled ECM–OCV–load model with quadratic per-step current solve and debounced cutoff, separating shutdown threshold from numerical feasibility.
- A calibration workflow that fuses public temperature–capacity data [5], NASA aging measurements [4], Zenodo OCV–SOC data [2] and ADB-derived component power coefficients.
- Cross-scenario TTE predictions and an elasticity-based sensitivity analysis that ranks the drivers most worth optimising in OS scheduling and user behaviour.

Fig. 1 summarises the pipeline. The remainder of the paper presents the model (Section II), the calibration data (Section III), the results and sensitivity analysis (Section IV), and discussion (Section V).

II. SYSTEM MODEL

A. State-of-Charge Dynamics

Let $z(t) \in [0, 1]$ be the battery State-of-Charge (SOC). Coulomb counting gives

$$\frac{dz}{dt} = -\frac{I(t)}{Q_{\max, \text{eff}}(T, N)}, \quad (1)$$

where $I(t)$ is the battery current (discharge positive), and the effective capacity factorises into a slowly varying capacity state, a temperature multiplier and an aging multiplier:

$$Q_{\max, \text{eff}}(T, N) = Q_{\max}(t) \phi_{\text{temp}}(T) \phi_{\text{age}}(N, T). \quad (2)$$

B. Second-Order Thevenin ECM

The terminal voltage uses ohmic resistance plus two polarisation RC branches [1]:

$$V_t(t) = V_{oc}(z(t)) - R_0(T) I(t) - v_1(t) - v_2(t), \quad (3)$$

$$\dot{v}_1 = -\frac{1}{R_1 C_1} v_1 + \frac{1}{C_1} I, \quad (4)$$

$$\dot{v}_2 = -\frac{1}{R_2 C_2} v_2 + \frac{1}{C_2} I, \quad (5)$$

where $V_{oc}(\cdot)$ is taken from the OCV-SOC lookup of Section III.

C. Constant-Power Load and Implicit Current Solve

The workload trace provides battery-side load power demand $P_{\text{load}}(t)$. With DC-DC efficiency η_{dc} , current and voltage are coupled by

$$P_{\text{load}}(t) = \eta_{\text{dc}} V_t(t) I(t). \quad (6)$$

Because V_t depends on I , substituting the ECM voltage produces a quadratic in I at each time step:

$$\eta_{\text{dc}} R_0(T) I^2 - \eta_{\text{dc}} (V_{oc}(z) - v_1 - v_2) I + P_{\text{load}} = 0, \quad (7)$$

and we keep the physically admissible (nonnegative) root. When the discriminant becomes infeasible at very low voltage we enforce a numerical lower bound $V_{\min}$ on the effective driving voltage. We deliberately separate this from the user-facing shutdown cutoff V_{cut} , used only to define TTE.

D. Debounced Cutoff

Short voltage dips during sustained loads can cause premature termination. We optionally apply a first-order low-pass on the terminal voltage,

$$\dot{v}_f = \frac{V_t - v_f}{\tau_f}, \quad (8)$$

and declare end-of-discharge when $v_f \leq V_{\text{cut}}$ instead of $V_t \leq V_{\text{cut}}$.

E. Power Consumption Model

Total load power is decomposed by component:

$$P_{\text{load}}(t) = P_{\text{base}} + \sum_k \alpha_k(t) x_k(t), \quad (9)$$

where $x_k(t)$ encodes component-specific usage (e.g., screen brightness, CPU utilisation, radio activity, GPS state) and $\alpha_k(t)$ is the corresponding power coefficient. This modular form lets us swap workload profiles without changing the battery side of the model.

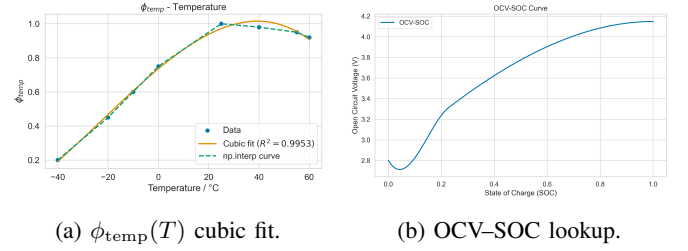


Fig. 2: Calibrated capacity multiplier and voltage characteristic.

F. Temperature and Aging

The capacity multiplier is a clipped cubic:

$$\phi_{\text{temp}}(T) = \text{clip}(\text{poly}_3(T), \phi_{\min}, \phi_{\max}), \quad (10)$$

and ohmic resistance follows an Arrhenius law [3]

$$R_0(T) = R_{0, \text{ref}} \exp\left(\beta \left(\frac{1}{T+273.15} - \frac{1}{298.15}\right)\right). \quad (11)$$

A cell-level energy balance [7] gives the thermal state

$$mc_p \frac{dT_b}{dt} = Q_{\text{gen}} - hA(T_b - T_{\text{amb}}), \quad (12)$$

with $Q_{\text{gen}} = I^2 R_0 + v_1^2 / R_1 + v_2^2 / R_2$. Aging enters multiplicatively via $\phi_{\text{age}}(N, T) = \exp(-k(T)N)$, with $k(T)$ fitted from cycle data.

III. CALIBRATION DATA

A. Temperature-Capacity Multiplier

We normalise reported capacity [5] by the value at 25 °C and fit a cubic

$$\begin{aligned} \phi_{\text{temp}}(T) = & -1.234 \times 10^{-6} T^3 - 8.259 \times 10^{-5} T^2 \\ & + 1.225 \times 10^{-2} T + 7.359 \times 10^{-1}. \end{aligned} \quad (13)$$

The fit (Fig. 2a) captures the well-known capacity drop at low temperatures and the saturation near room conditions.

B. OCV-SOC Curve

We construct the OCV-SOC lookup from the Zenodo cell dataset [2] after smoothing and monotone resampling (Fig. 2b). It is consumed as a 1-D table during simulation.

C. Cycle Aging

Fitting an exponential capacity-fade law to the NASA aging set [4] gives a temperature-dependent aging coefficient

$$k(T) = 3.7 \times 10^{-5} T + 1.4 \times 10^{-3}, \quad (14)$$

i.e. degradation accelerates monotonically with T , consistent with Arrhenius-type kinetics.

TABLE I: Estimated per-unit component power coefficients.

Comp.	Base	Screen	CPU	WiFi	Cell.	GPS
α_k	0.45	2.0	2.5	0.8	1.2	0.8
Comp.	BT	Audio	Flash	RAM	PwrSv	
α_k	0.15	0.35	1.5	0.3	0.25	

TABLE II: Component activity scaling under four representative usage profiles.

Profile	Scrn	CPU	WiFi	Cell.	GPS	Audio
Idle	0.0	0.8	0.4	0.6	0.8	0.25
Video	1.6	1.5	0.6	0.8	0.2	0.80
Gaming	1.8	2.2	0.5	0.7	0.1	0.20
Mixed	1.0	1.4	0.5	0.7	0.4	0.40

D. Component Power Coefficients

We extract system-level traces from a Xiaomi 14 Pro using ADB `batterystats`, then estimate baseline coefficients by linear regression against component activity vectors. Table I reports the resulting per-unit coefficients (W per unit activity), and Table II their scenario-specific scalings. Stochastic perturbations are added at runtime to emulate OS scheduling jitter while preserving scenario means; an example is shown in Fig. 3.

IV. RESULTS AND SENSITIVITY ANALYSIS

We simulate a fully charged 4724 mAh cell at 25 °C with $V_{\text{cut}} = 3.3$ V. The implicit current solve in (7) runs at 1 Hz; the debounced cutoff uses $\tau_f = 30$ s.

A. Time-to-Empty Across Profiles

Fig. 4 reports SOC trajectories and the resulting TTE values. The model predicts 39.45 h (idle), 9.06 h (video), 5.34 h (gaming) and 4.08 h (mixed): close to an order-of-magnitude spread between idle and the heaviest mixed-use profile. The TTE ordering matches qualitative expectations: gaming combines high CPU and bright display, while *mixed* adds sustained network activity.

B. Component-Resolved Drain Attribution

Integrating the simulated power traces by component yields the energy attribution shown in Fig. 5. Idle drain is dominated by baseline power; in mixed daily use, cellular and WiFi together account for roughly half the energy budget; gaming and video are CPU- and screen-heavy, with audio rising in multimedia scenarios.

C. Temperature and Initial-SOC Effects

Fig. 6 shows TTE under the mixed profile across ambient temperatures, and Fig. 7 extends this to a TTE surface over initial SOC and temperature. Cold conditions sharply compress TTE through reduced effective capacity. Slightly elevated temperatures can extend short-term TTE by lowering ohmic resistance, even though they accelerate long-term degradation. The widely held view that “heat shortens battery life” therefore conflates instantaneous and lifetime effects.

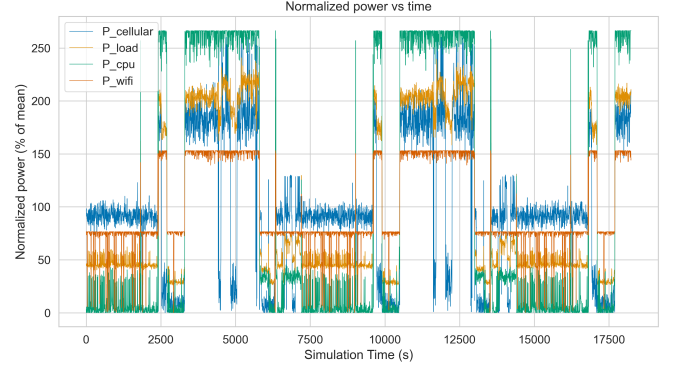


Fig. 3: Stochastic component-power realisations around scenario baselines emulate OS scheduling variability.

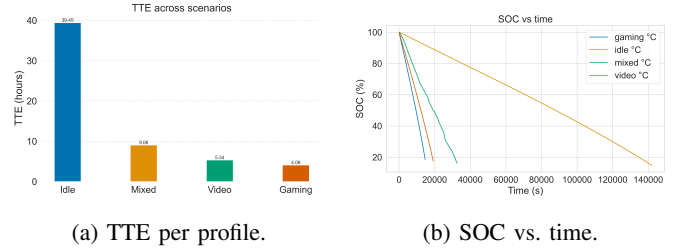


Fig. 4: TTE and SOC trajectories across the four usage profiles at 25 °C.

D. Sensitivity Analysis

For each multiplicative parameter p we compute the dimensionless TTE elasticity

$$S_p = \frac{\partial \ln \text{TTE}}{\partial \ln p} = \frac{p}{\text{TTE}} \frac{\partial \text{TTE}}{\partial p}. \quad (15)$$

Fig. 8 ranks S_p across profiles. The dominant levers are nominal capacity, screen demand, network activity, CPU load and baseline draw; flashlight, Bluetooth and RAM rank consistently low. A complementary $\pm 10\%$ tornado perturbation under the gaming profile (Fig. 9) reaches the same conclusion through a more direct sensitivity, with screen and CPU producing the largest TTE swings.

V. DISCUSSION

A. Implications for OS Scheduling

The sensitivity ranking maps directly to OS-level levers:

- Reduce persistent system services and periodic wake-ups to lower the baseline.
- Tighten background-execution and network policies; cellular tail energy and repeated WiFi handshakes inflate the network share most visibly in idle and mixed scenarios.
- Use energy-aware DVFS and core selection so that boosting is reserved for latency-critical work.
- Batch network requests and coordinate cellular/WiFi switching to amortise radio wake costs.

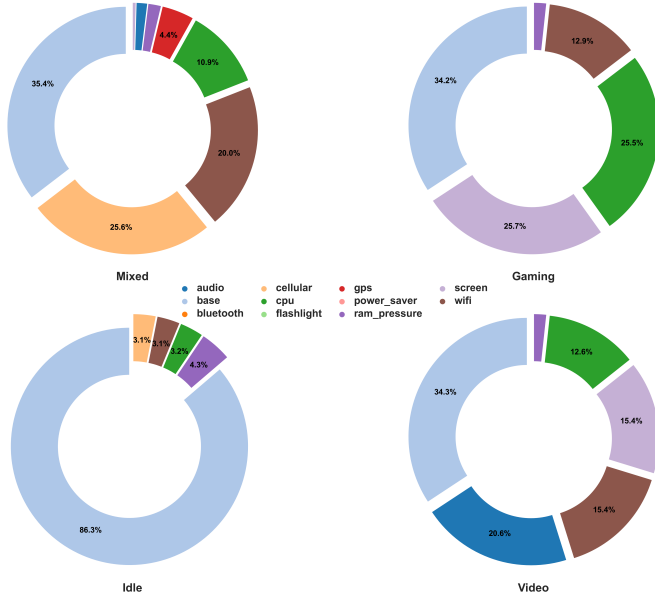


Fig. 5: Component-wise share of total energy across the four profiles.

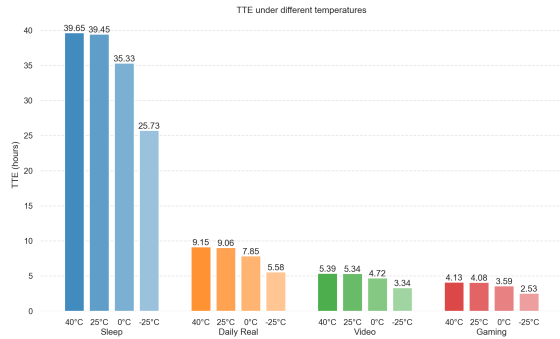


Fig. 6: TTE vs. ambient temperature (mixed profile).

B. User-Facing Recommendations

Translating the same ranking to end users:

- Lower screen brightness or enable adaptive brightness, and shorten screen timeout; the display is consistently in the top three drain contributors.
- Restrict non-essential auto-start and background apps that keep CPU and radios active.
- Avoid running both WiFi and cellular when not needed, especially in poor-signal environments.
- Limit sustained heavy gaming or high-bitrate streaming, both for instantaneous drain and for thermal-stress-driven aging.

C. Limitations

Component coefficients are calibrated from a single device; thermal effects are lumped, and the random fluctuation model is not learned from large-scale traces. Generalising to other consumer devices (wearables, AR/VR headsets) is straightfor-

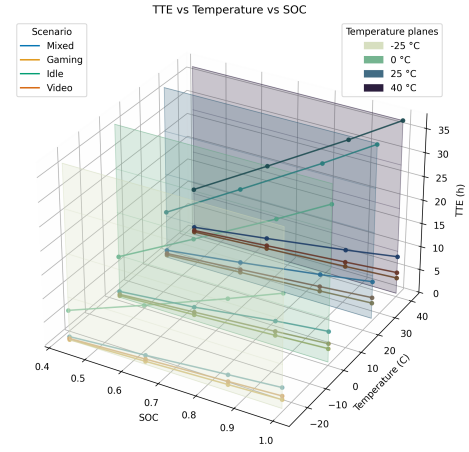


Fig. 7: TTE surface over initial SOC and ambient temperature.

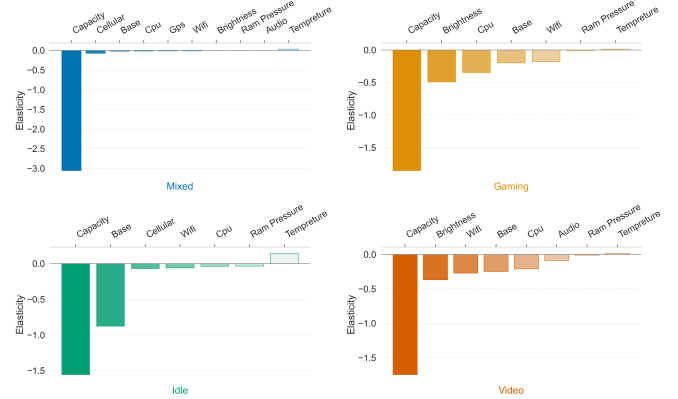


Fig. 8: TTE elasticity S_p across usage profiles.

ward in form but requires per-platform calibration of α_k and the ECM parameters.

VI. CONCLUSION

We presented a continuous-time, ECM-OCV battery model with constant-power load coupling and a debounced cutoff, calibrated from public datasets and on-device ADB traces. Across four representative smartphone workloads the model yields physically interpretable TTE predictions and a clear ranking of optimisation levers: capacity, screen, network, CPU and baseline. The pipeline is reproducible from public data and portable to other consumer-electronics platforms, and provides a quantitative basis for both OS-level scheduling decisions and user-facing battery-life guidance.

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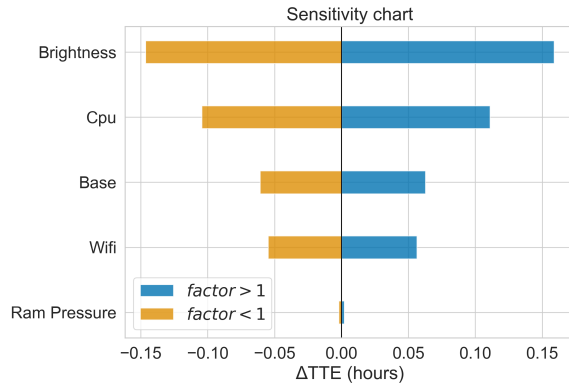


Fig. 9: Tornado plot: $\pm 10\%$ perturbations on per-component coefficients under the gaming profile.

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